

MACHINE LEARNING FOR SOFTWARE CONSTRUCTION

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WHAT IS MACHINE LEARNING?

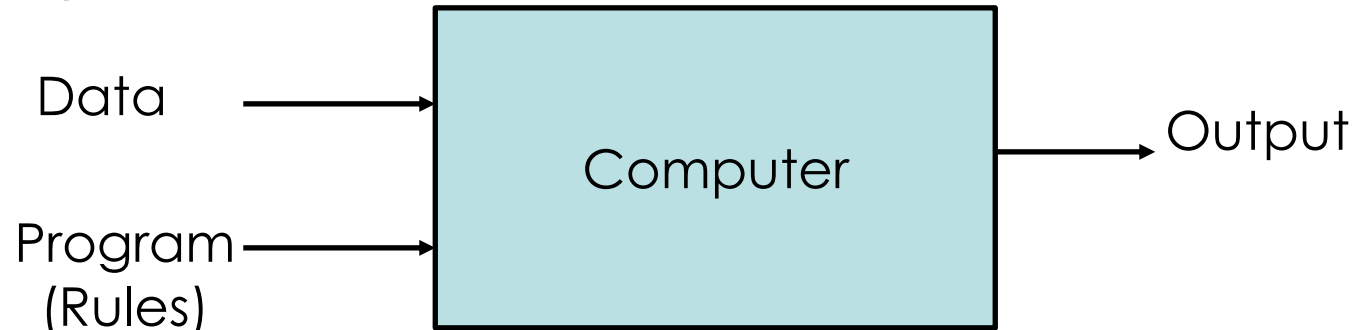
“ Field of study that gives computers the ability to learn without being explicitly programmed. ”

— Arthur Samuel

PROGRAMMING VS ML

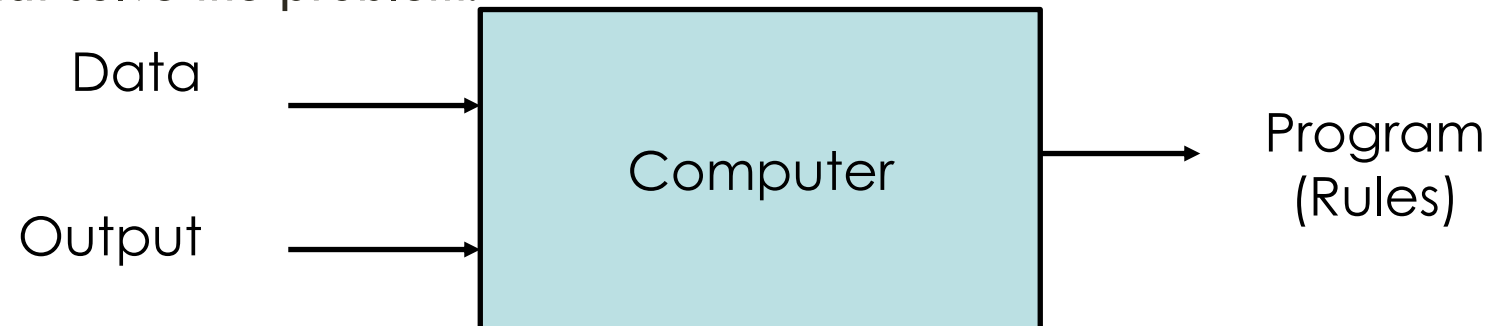
➤ Traditional Programming

- You have the data and you are a programmer writing a program on the computer to produce the output



➤ Machine Learning

- You have the data and desired output and would like to train the computer to produce the program that solve the problem.



ACTIVITY RECOGNITION



```
if(speed<4){  
    status=WALKING;  
}
```



```
if(speed<4){  
    status=WALKING;  
} else {  
    status=RUNNING;  
}
```



```
if(speed<4){  
    status=WALKING;  
} else if(speed<12){  
    status=RUNNING;  
} else {  
    status=BIKING;  
}
```



```
// Uh oh
```

ACTIVITY RECOGNITION



```
0101001010100101
0101001010101001
0111010100101010
0101010010101001
010100101010
```

Label =
WALKING



```
1010100101001010
1010101010010010
0100010010011111
0101011111010100
100111101011
```

Label =
RUNNING



```
1001010011111010
1011101010111010
1011101010101111
0101010111111110
001111010101
```

Label =
BIKING



```
1111111111010011
1010011111010111
1101010101110101
0101011101010101
010100111110
```

Label = GOLFING
(Sort of)

WHAT IS MACHINE LEARNING?

- Let's say you want to solve Character Recognition
- Hard way: Understand handwriting/characters
- Lazy way: Throw data!

س ه و أ ك ل و د س ش ع ف
ع ص ح م ه و ل ل غ ه ت ج
ه ش د ح م ع ر ط أ ي
إ ص ص ع غ ه ش ش ش د و ر
ف ل ك ن ق ز ذ ن ع ب غ
ت ن ع م ض ط ك ل م ل ل ل
ق ح ص ح ح خ ر ص ع ف ه
أ ن ي س أ ث خ ذ ط ظ ل
س ش ش ع ن ن س ط ه
ك خ أ ك ج ج س ش ش
ب ت ت ت ن و ر د ذ

WHAT IS MACHINE LEARNING?

➤ [Arthur Samuel, 1959]

- Field of study that gives computers the ability to learn without being explicitly programmed

➤ [Kevin Murphy] algorithms that

- automatically detect patterns in data
- use the uncovered patterns to predict future data or other outcomes of interest

➤ [Tom Mitchell] algorithms that

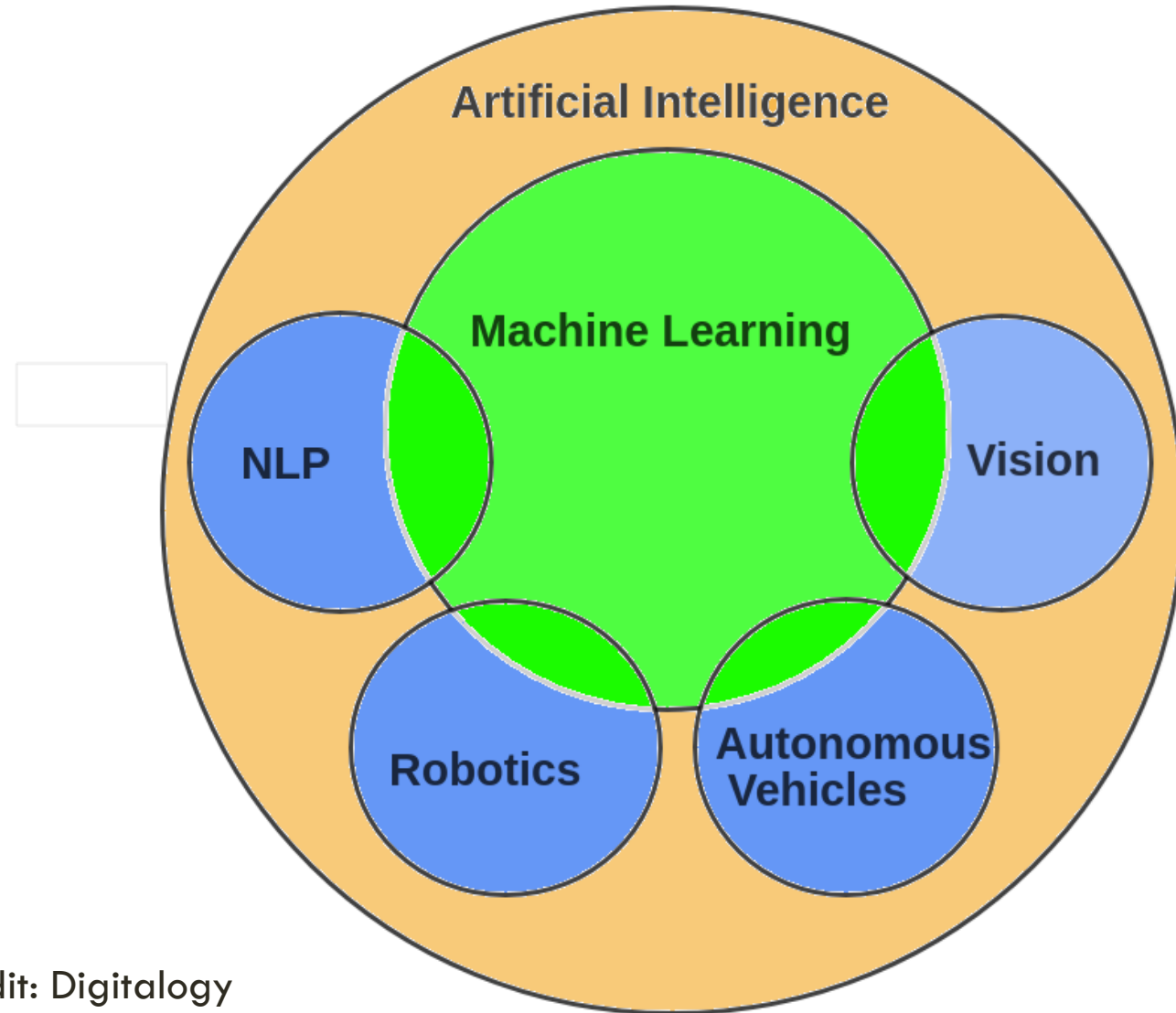
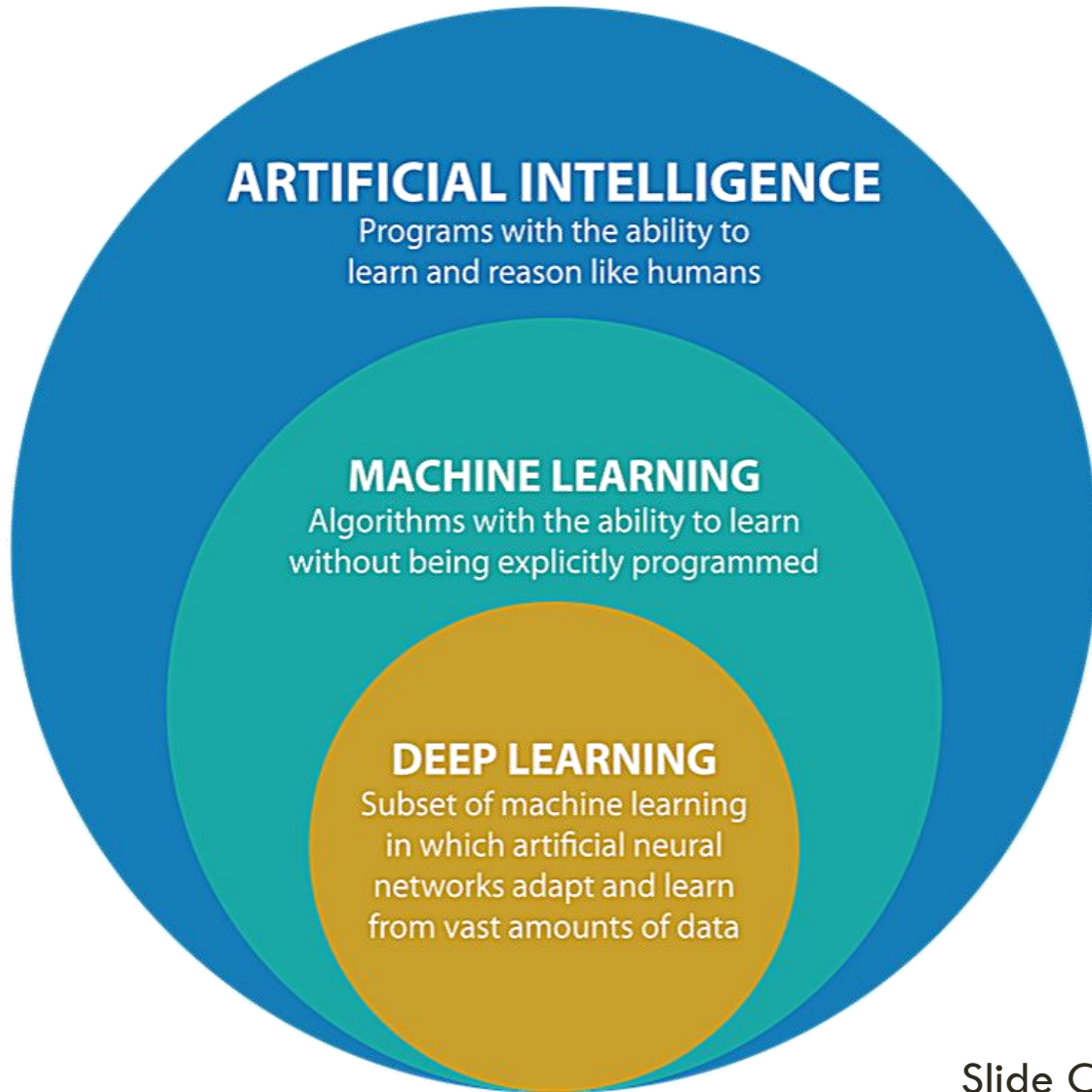
- improve their performance (P)
- at some task (T)
- with experience (E)

EXAMPLES

➤ Text Classification

- Task T
 - Assigning texts to a set of predefined categories
- Performance measure P
 - percent of texts correctly classified
- Training experiences E
 - A database of texts with their corresponding categories

AI, ML, AND DL



TYPES OF LEARNING



TYPES OF LEARNING

➤ **Supervised learning**

- Training data includes desired outputs

➤ **Unsupervised learning**

- Training data does not include desired outputs

➤ **Weakly or Semi-supervised learning**

- Training data includes a few desired outputs

➤ **Reinforcement learning**

- Rewards from sequence of actions

SUPERVISED LEARNING

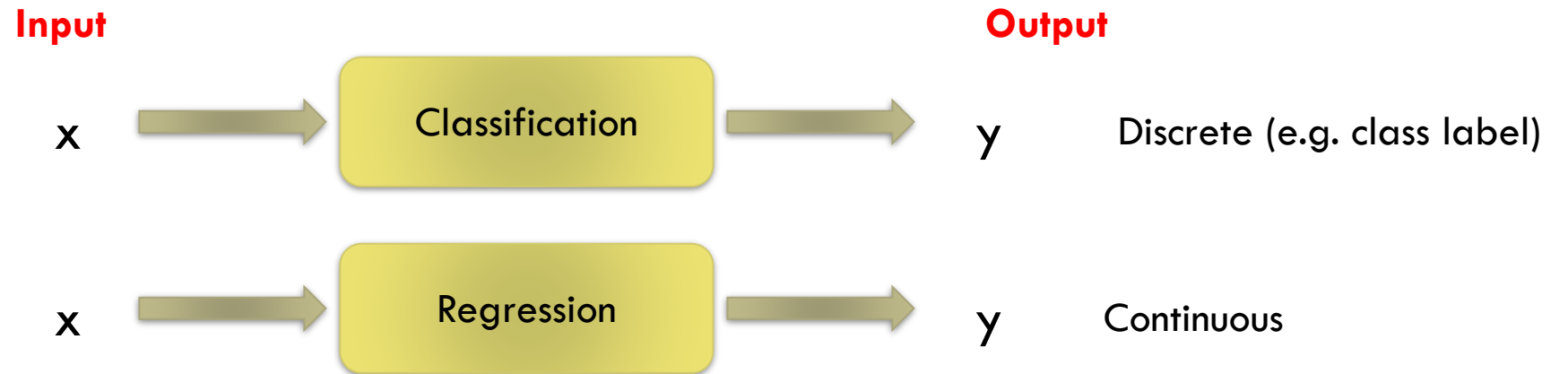
- Training data provided as pairs (x,y)
- The goal is to predict an “output” y from an “input” x
- Output y for each input x is the “supervision” that is given to the learning algorithm.
 - Often obtained by manual “annotation” of the inputs x
 - Can be costly to do
- Most common examples
 - Classification
 - Regression

UNSUPERVISED LEARNING

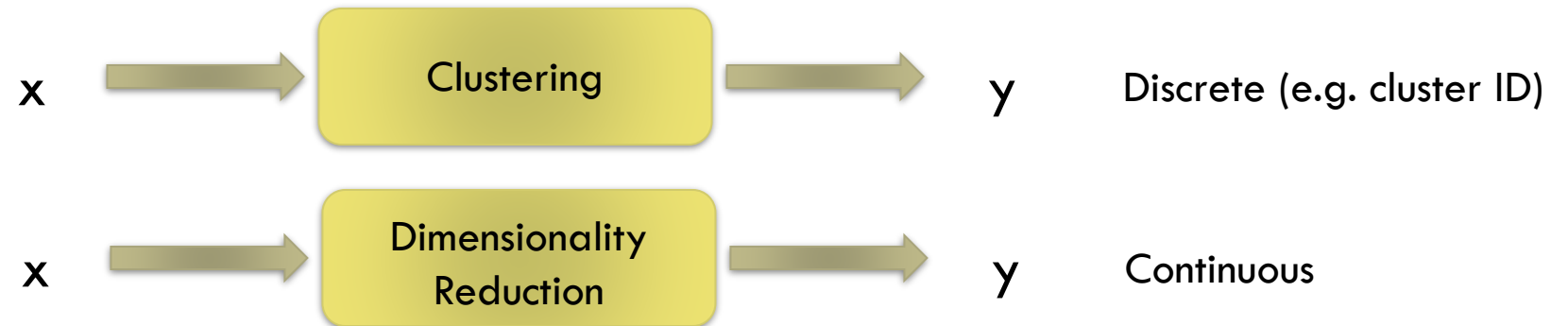
- Input data x given without desired output variables y .
- Goals is to learn something about the “structure” of the data
- **Examples include**
 - Clustering
 - Dimensionality reduction
 - Anomaly detection
 - Feature extraction

TASKS

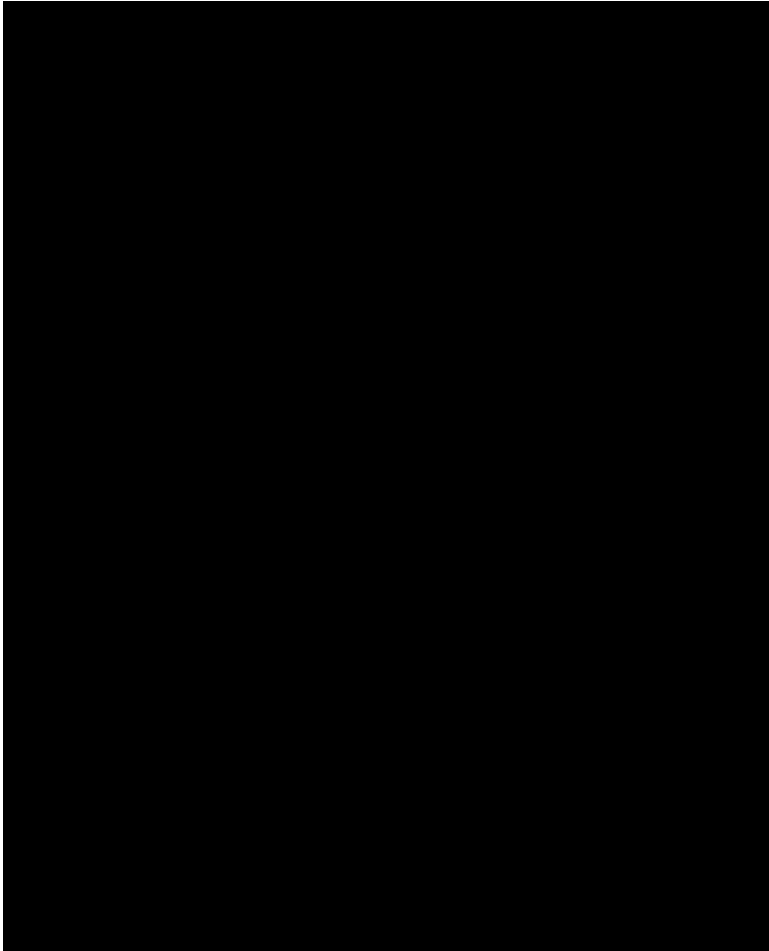
Supervised Learning



Unsupervised Learning



REINFORCEMENT LEARNING



- An **agent** interacts with an **environment** (e.g. Atari Breakout game)
- In each time step,
 - the agent receives **observations** (e.g. pixels) which give it information about the state (e.g. positions of the ball and paddle)
 - the agent picks an **action** (e.g. keystrokes) which affects the state
- The agent periodically receives a **reward** (e.g. points)
- The agent wants to learn a **policy**, or mapping from observations to actions, which maximizes its average reward over time

BUILDING MACHINE LEARNING SYSTEM

STEPS IN MACHINE LEARNING

- Data collection
 - “training data”, optionally with “labels” provided by a “teacher”.
- Representation (Features Extraction)
 - how the data are encoded into “features” when presented to learning algorithm.
- Modeling (Classifier)
 - choose the class of models that the learning algorithm will choose from.
- Estimation (Training)
 - find the model that best explains the data: simple and fits well.
- Validation
 - evaluate the learned model and compare to solution found using other model classes.
- Apply learned model to new “test” data

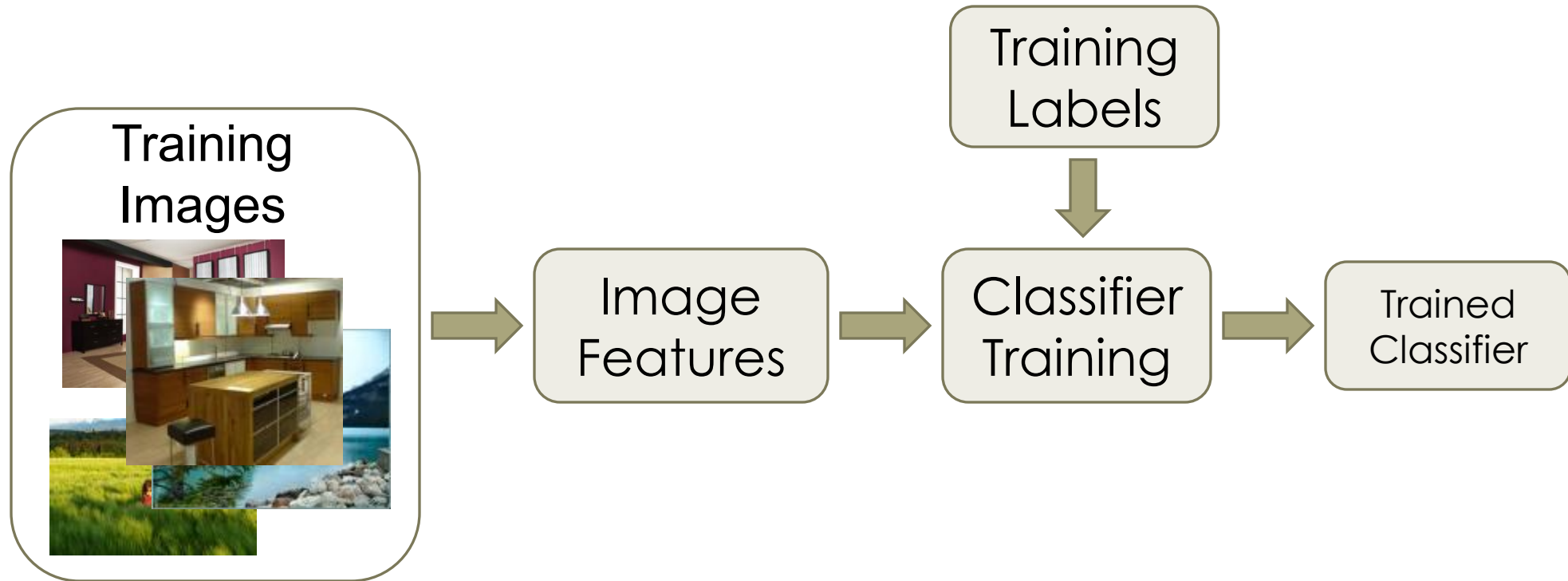
ML IN A NUTSHELL

- Tens of thousands of machine learning algorithms
 - Hundreds new every year
- Decades of ML research oversimplified:
 - All of Machine Learning:
 - Learn a mapping from input to output $f: X \rightarrow Y$
 - X : emails, Y : {spam, notspam}

ML IN A NUTSHELL

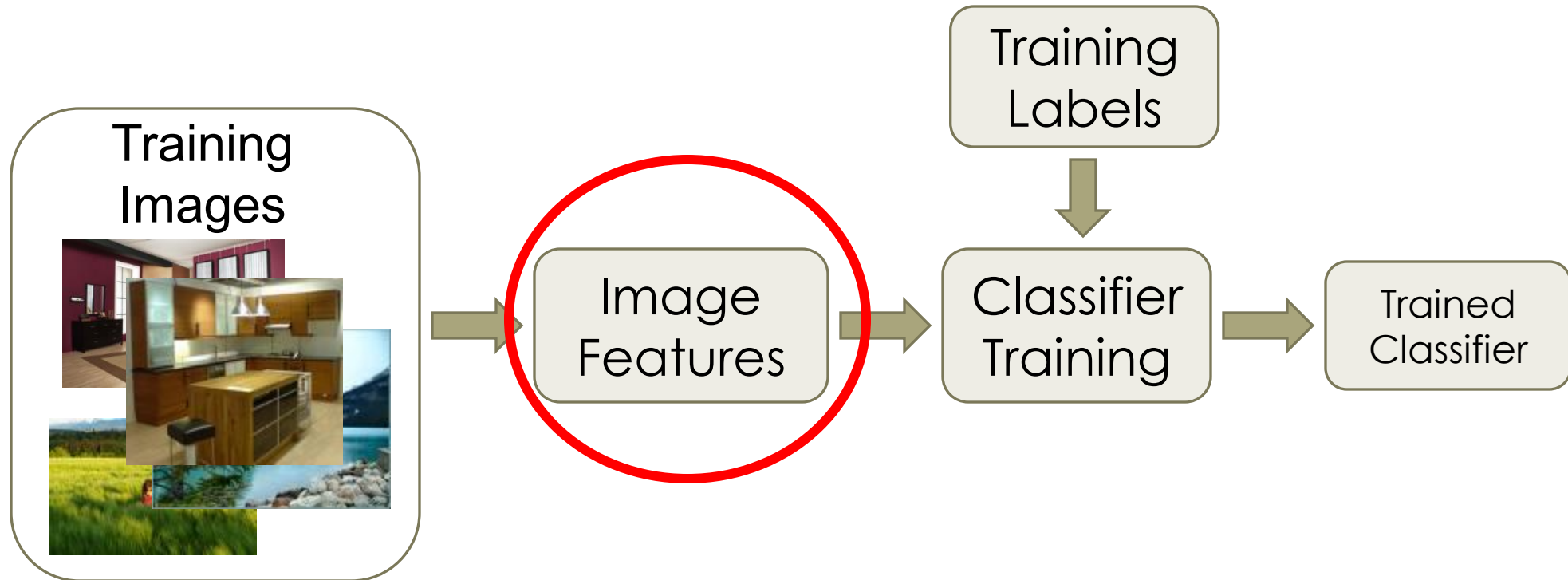
- Input: x (images, text, emails...)
- Output: y (spam or non-spam...)
- (Unknown) Target Function
 - $f: X \rightarrow Y$ (the “true” mapping / reality)
- Data
 - $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$
- Model / Hypothesis Class
 - $g: X \rightarrow Y$
 - $y = g(x) = \text{sign}(w^T x)$

IMAGE CATEGORIZATION



Training set (seen data) is used to build the model (determine its parameters)

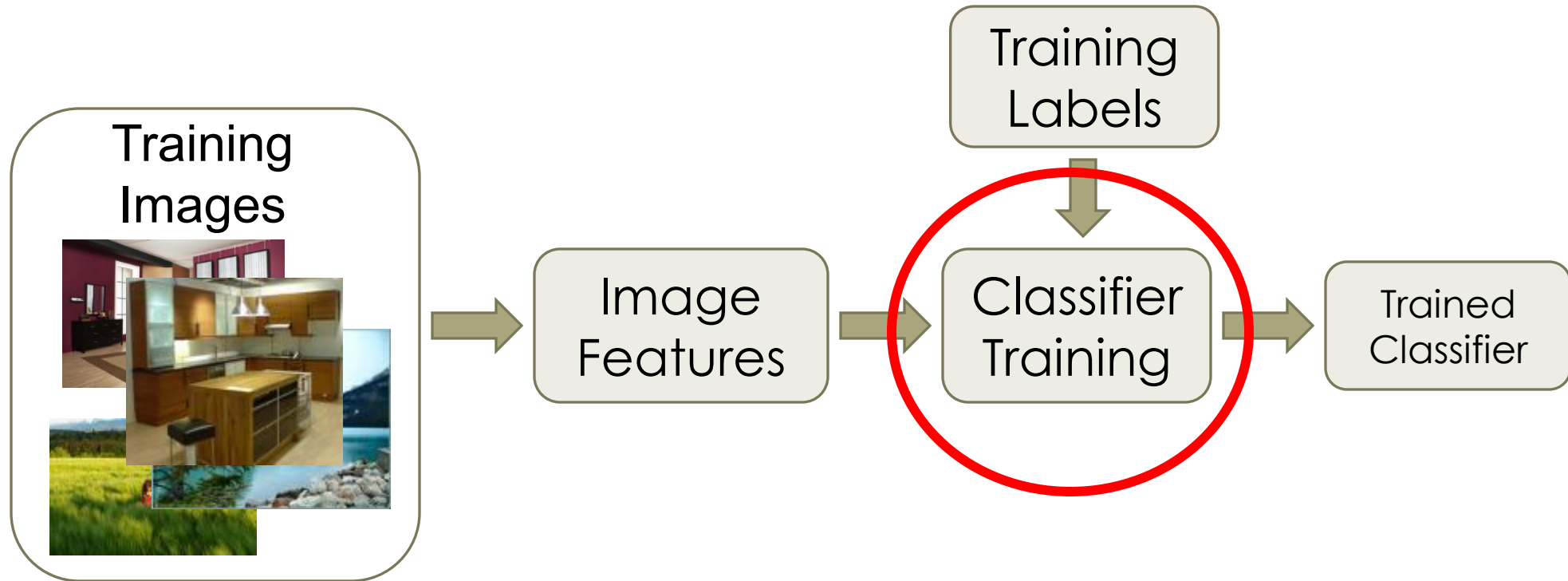
IMAGE FEATURES



GENERAL PRINCIPLES OF REPRESENTATION

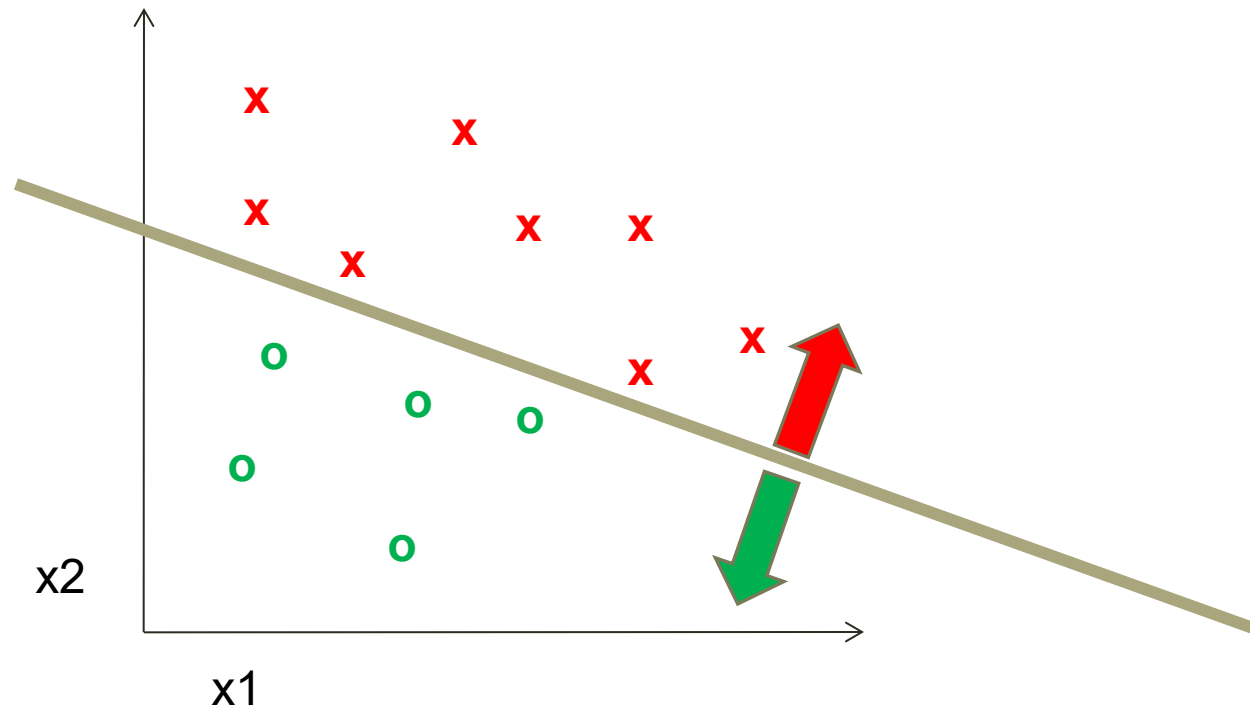
- **Robust** (i.e., they should normally be invariant to translation, orientation (rotation), scale, and illumination a
- **Discriminating** (i.e., the range of values for objects in different classes should be different and preferably be well separated and non-overlapping)
- **Reliable** (i.e., all objects of the same class should have similar values)
- **Independent** (i.e., uncorrelated; as length and area are correlated and it would be wasteful to consider both as separate features)

CLASSIFIERS



LEARNING A CLASSIFIER

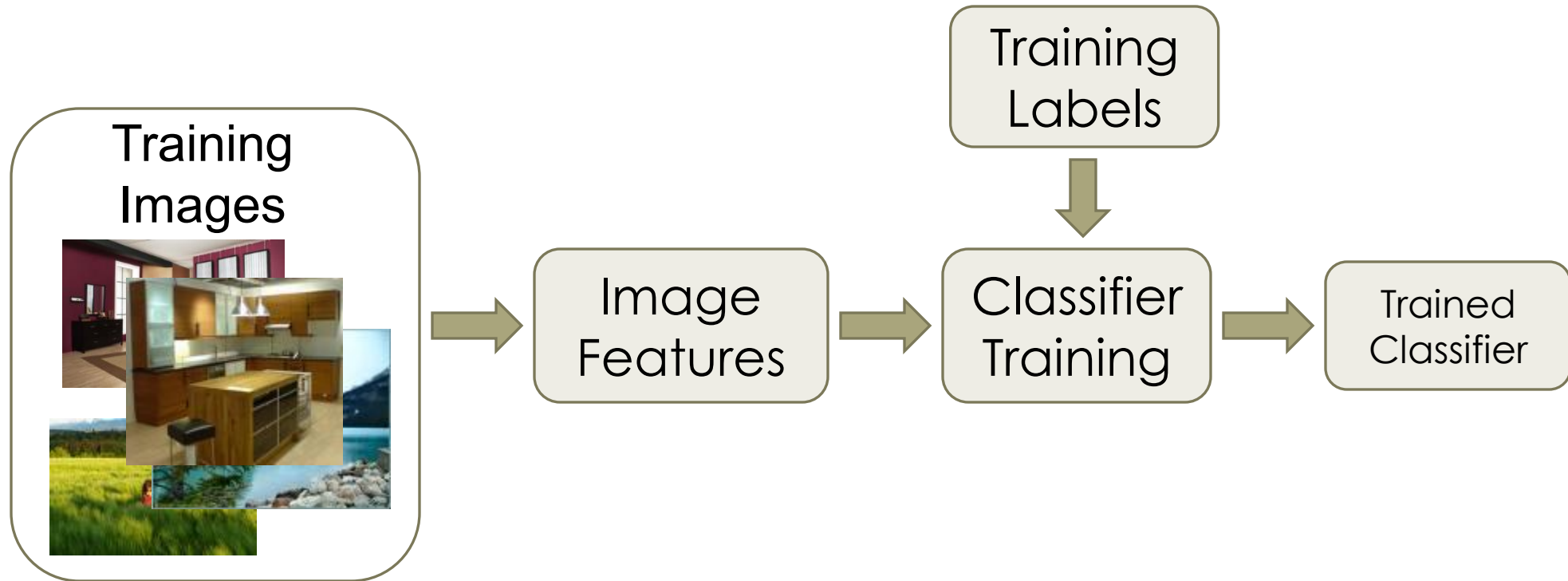
Given some set of features with corresponding labels, learn a function to predict the labels from the features



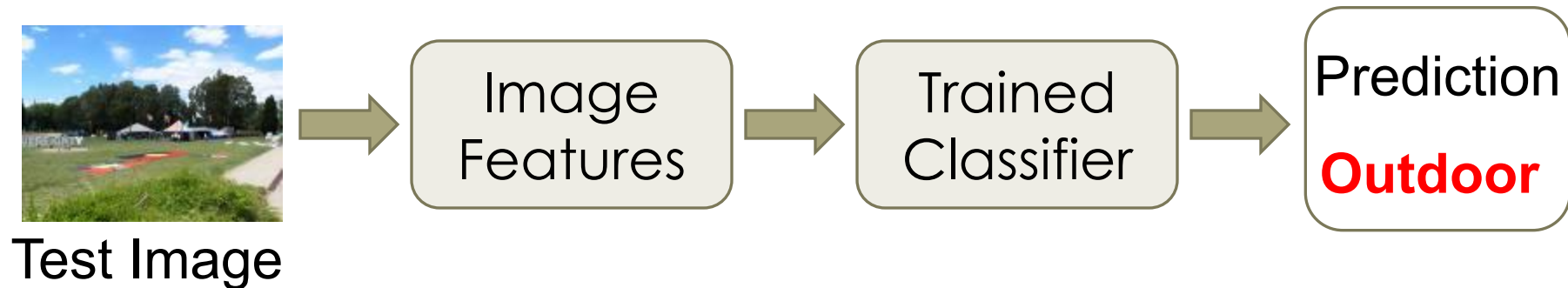
MODEL CLASS

- Neural networks (Deep learning)
- Nearest Neighbor Classifier
- Support vector machines
- Decision trees
- Random Forest
- Naive Bayes classifier
- Graphical models (Bayes/Markov nets)
- Ensembles Models (e.g. bagging, boosting)
- Regression analysis
- Etc.

IMAGE CATEGORIZATION



Testing



KNN MODEL



IMAGE CLASSIFICATION

Image Classification: a core task in Computer Vision



(assume given set of discrete labels)
{dog, cat, truck, plane, ...}



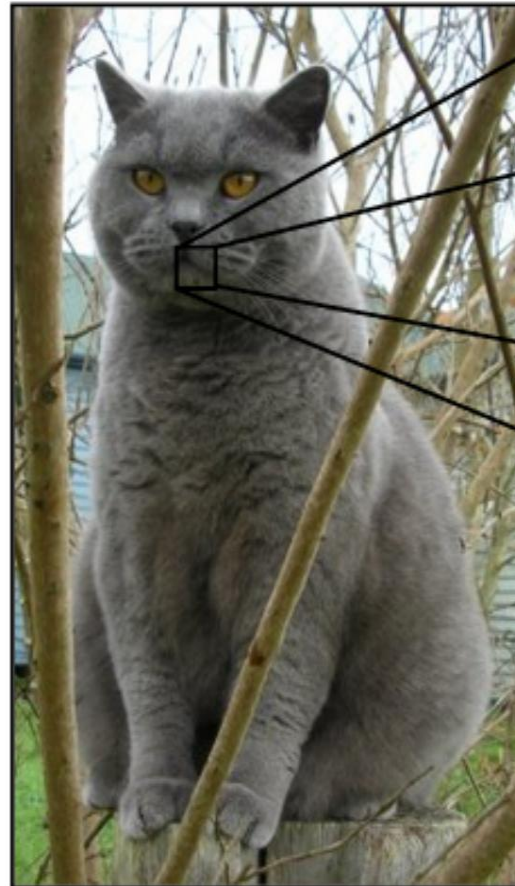
cat

IMAGE CLASSIFICATION

The problem:
semantic gap

Images are represented as R^d
arrays of numbers

- E.g. R^3 with integers
between $[0, 255]$, where
 $d=3$ represents 3 color
channels (RGB)



08	02	22	97	38	15	00	40	00	75	04	05	07	78	52	12	50	77	81	88
49	49	99	40	17	81	18	57	60	87	17	40	98	43	69	42	04	56	62	00
81	49	31	73	55	79	14	29	93	71	40	67	53	88	30	03	49	13	36	65
52	70	95	23	04	60	11	42	62	21	68	56	01	32	56	71	37	02	36	91
22	31	16	71	51	62	85	59	41	92	36	54	22	40	40	28	66	33	13	80
24	47	32	60	99	03	45	02	44	75	33	53	78	36	84	20	35	17	12	50
32	98	81	28	64	23	67	10	26	38	40	67	59	54	70	66	18	38	64	70
67	26	20	68	02	62	12	20	95	63	94	39	63	08	40	91	66	49	94	21
24	55	58	05	66	73	99	26	97	17	78	78	96	83	14	88	34	89	63	72
21	36	23	09	75	00	76	44	20	45	35	14	00	61	33	97	34	31	33	95
78	17	53	28	22	75	31	67	15	94	03	80	04	62	16	14	09	53	56	92
16	39	05	42	96	35	31	47	55	58	88	24	00	17	54	24	36	29	85	57
86	56	00	48	35	71	89	07	05	44	44	37	44	60	21	58	51	54	17	58
19	80	81	68	05	94	47	69	28	73	92	13	86	52	17	77	04	89	55	40
04	52	08	83	97	35	99	16	07	97	57	32	16	26	26	79	33	27	98	66
89	86	68	87	57	62	20	72	03	46	33	67	46	55	12	32	63	93	53	69
04	42	16	73	35	85	39	11	24	94	72	18	08	46	29	32	40	62	76	36
20	69	36	41	72	30	23	88	34	82	93	69	82	67	59	85	74	04	36	16
20	73	35	29	78	31	90	01	74	31	49	71	48	56	81	16	23	57	05	54
01	70	54	71	83	51	54	69	16	92	33	48	61	43	52	01	89	17	67	48

What the computer sees

CHALLENGES

Viewpoint variation



Scale variation



Deformation



Occlusion



Illumination conditions



Background clutter



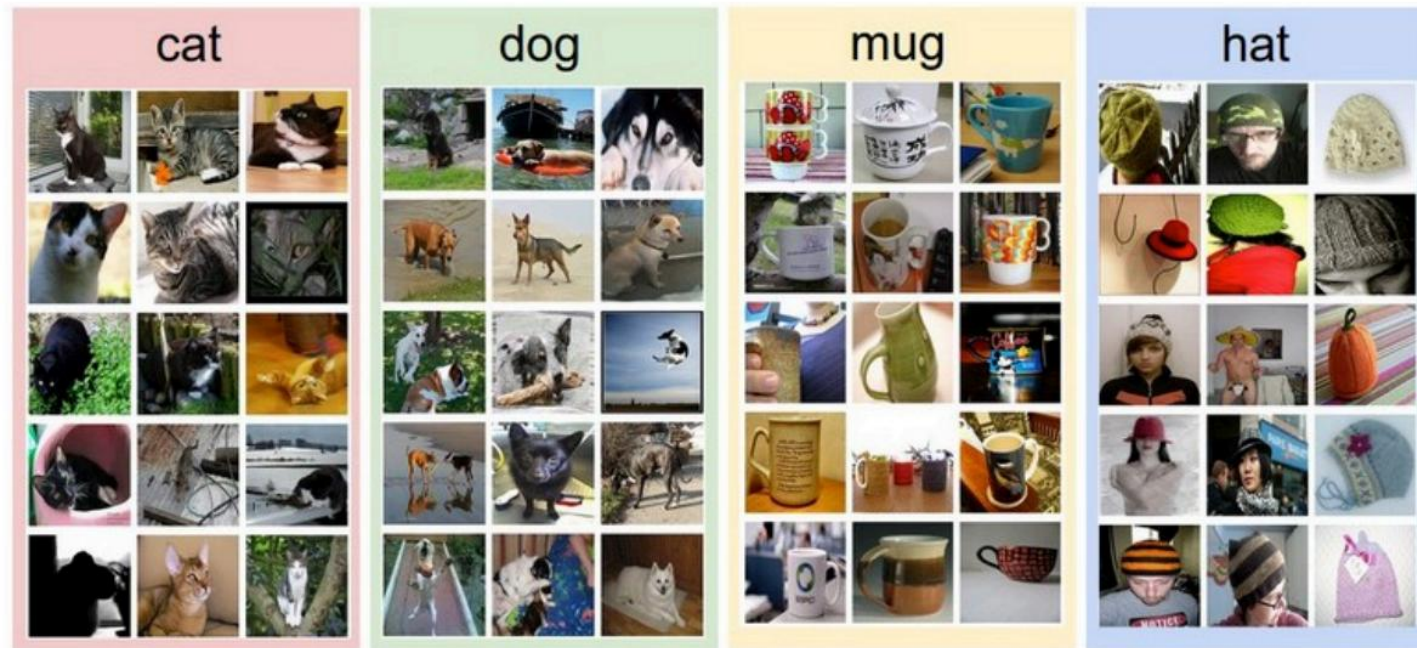
Intra-class variation



AN IMAGE CLASSIFIER

- Data-driven approach:
 1. Collect a dataset of images and label them
 2. Use Machine Learning to train an image classifier
 3. Evaluate the classifier on a withheld set of test images

Example training set



NEAREST NEIGHBOR CLASSIFIER

➤ First classifier: Nearest Neighbor Classifier

Function `train(train_images,train_labels)`



Remember all training images and their labels

Function `class_label=predict(image)`

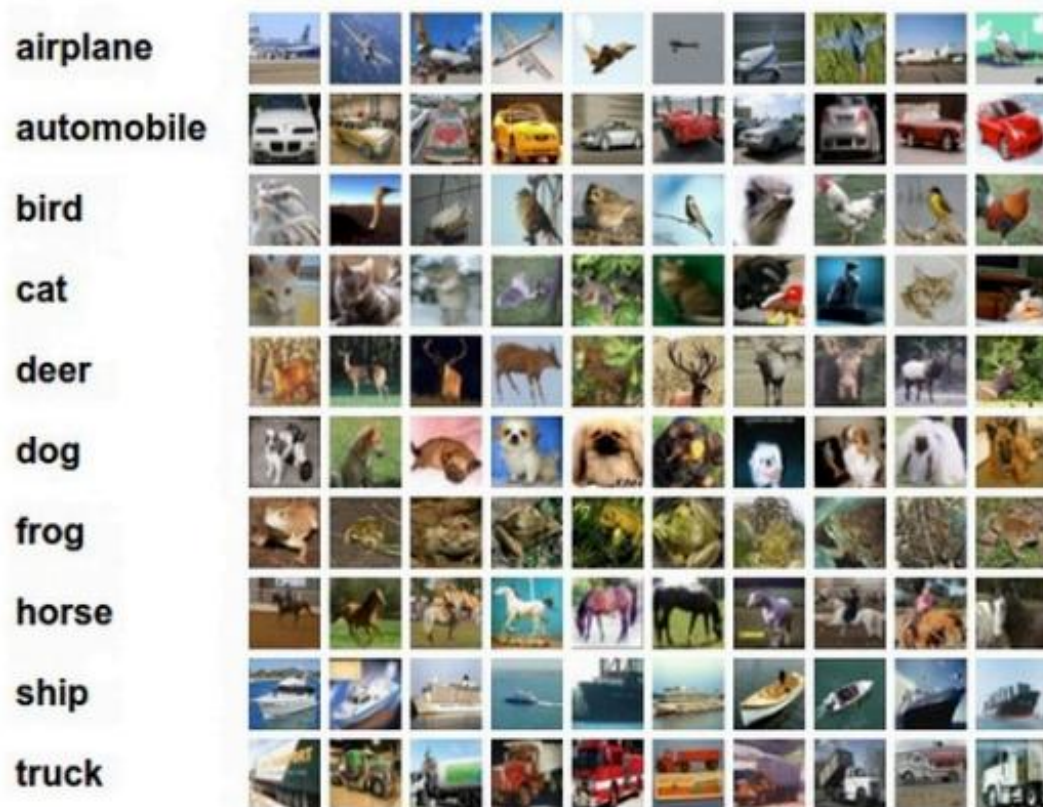


Predict the label of the most similar training image

NEAREST NEIGHBOR CLASSIFIER

- Example dataset: CIFAR-10
 - 10 labels
 - 50,000 training images
 - 10,000 test images.

For every test image (first column),
examples of nearest neighbors in rows



NEAREST NEIGHBOR CLASSIFIER

How do we compare the images? What is the **distance metric**?

L1 distance:

(Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

Where I_1 denotes image 1,
and p denotes each pixel

test image

56	32	10	18
90	23	128	133
24	26	178	200
2	0	255	220

training image

10	20	24	17
8	10	89	100
12	16	178	170
4	32	233	112

-

=

pixel-wise absolute value differences

46	12	14	1
82	13	39	33
12	10	0	30
2	32	22	108

→ 456

HYPERPARAMETERS

- The choice of distance is a **hyperparameter** common choices:

L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p (I_1^p - I_2^p)^2}$$

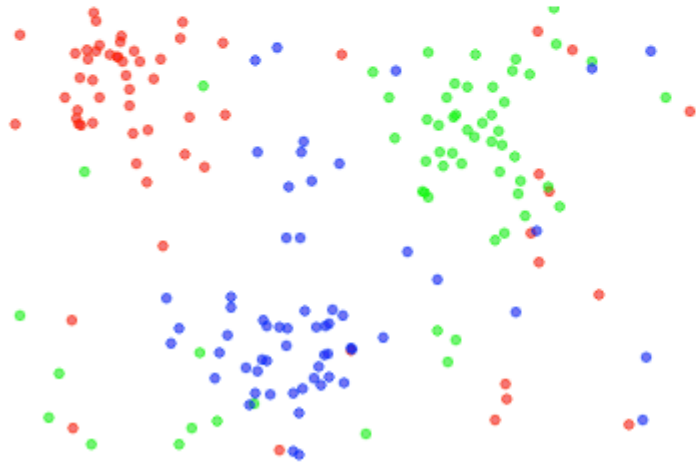
hyperparameters, is a kind of parameters that cannot be directly learned from the regular training process. These parameters express “higher-level” properties of the model such as its complexity or how fast it should learn. They are usually fixed before the actual training process begins.

K-NEAREST NEIGHBOR CLASSIFIER

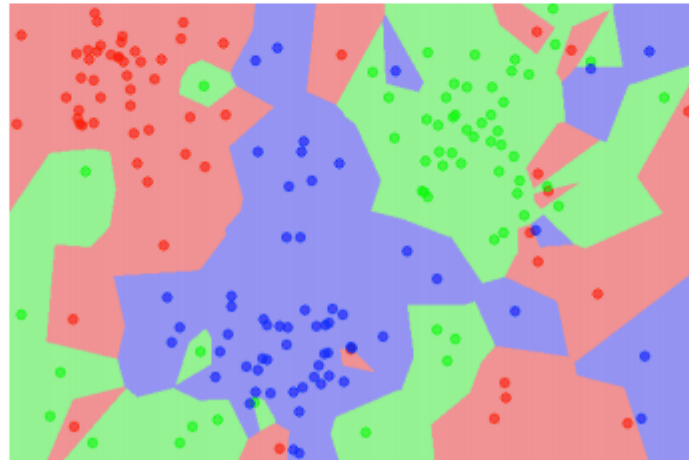
k-Nearest Neighbor

find the k nearest images, have them vote on the label

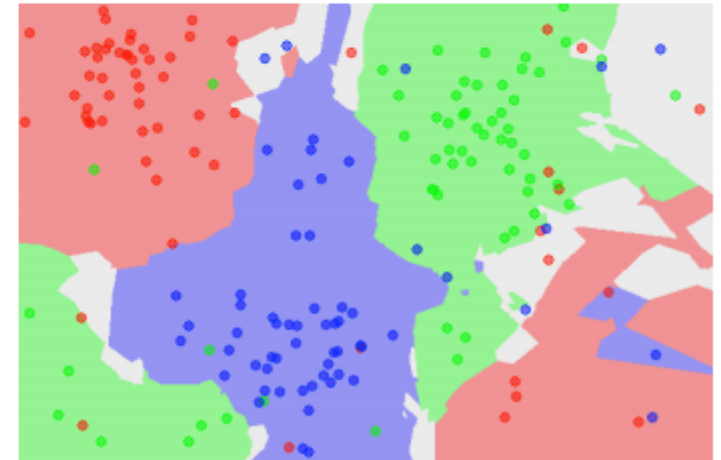
the data



NN classifier



5-NN classifier



HYPERPARAMETERS

What is the **best distance** to use?

What is the **best value of k** to use?

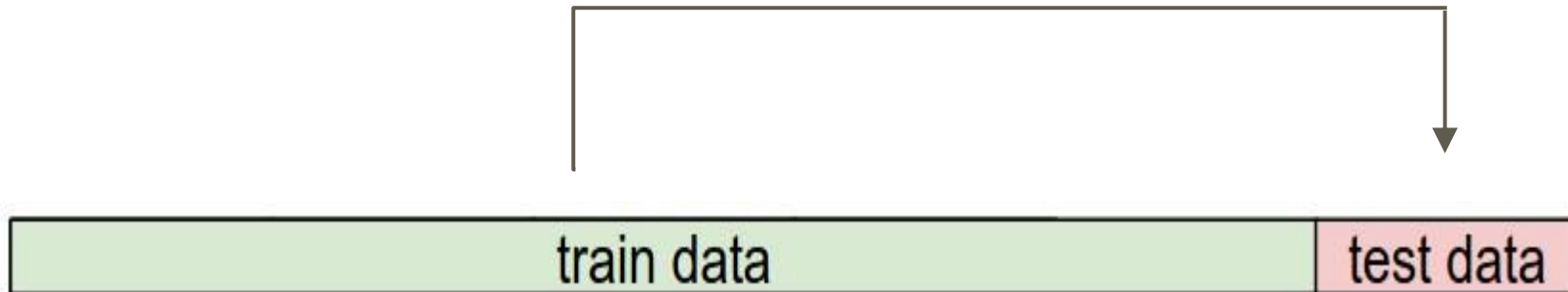
i.e. how do we set the hyperparameters?

Very problem-dependent.

Must try them all out and see what works best.

HYPERPARAMETERS

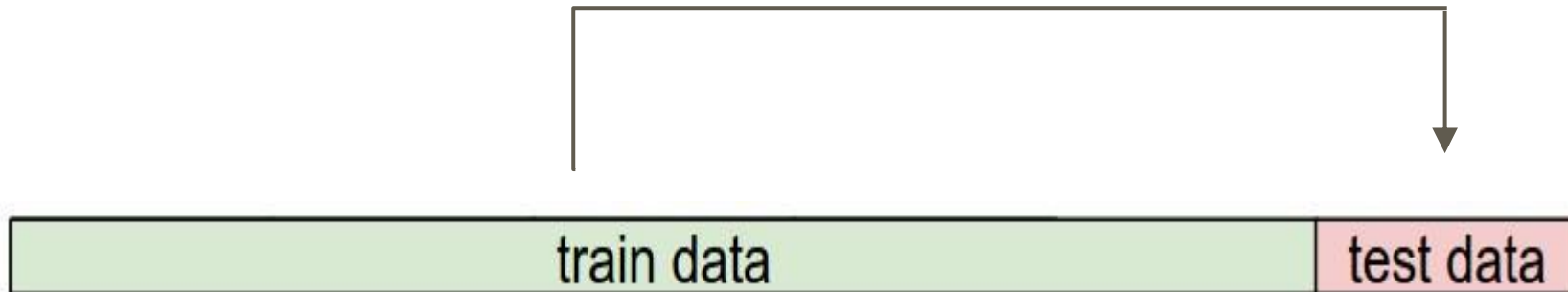
Try out what hyperparameters work best on test set.



HYPERPARAMETERS

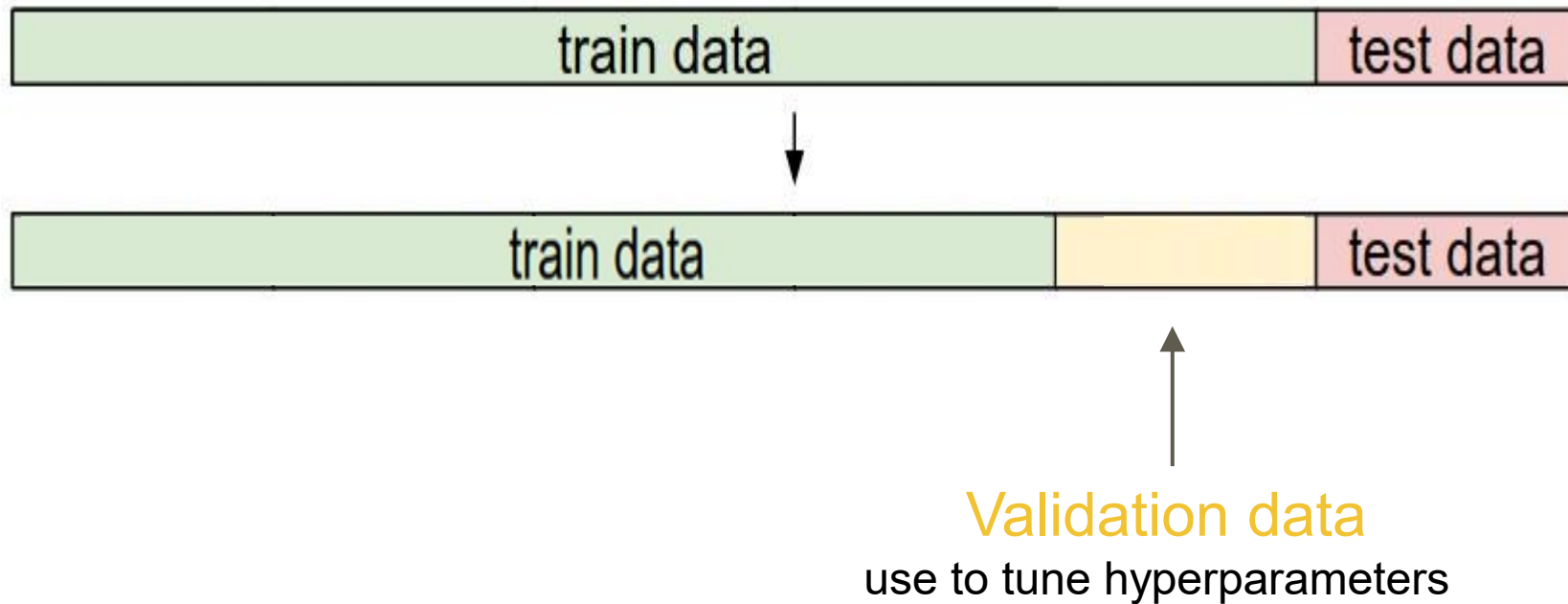
Trying out what hyperparameters work best on test set:

Very bad idea. The test set is a proxy for the generalization performance!
Use only **Once**, at the end.



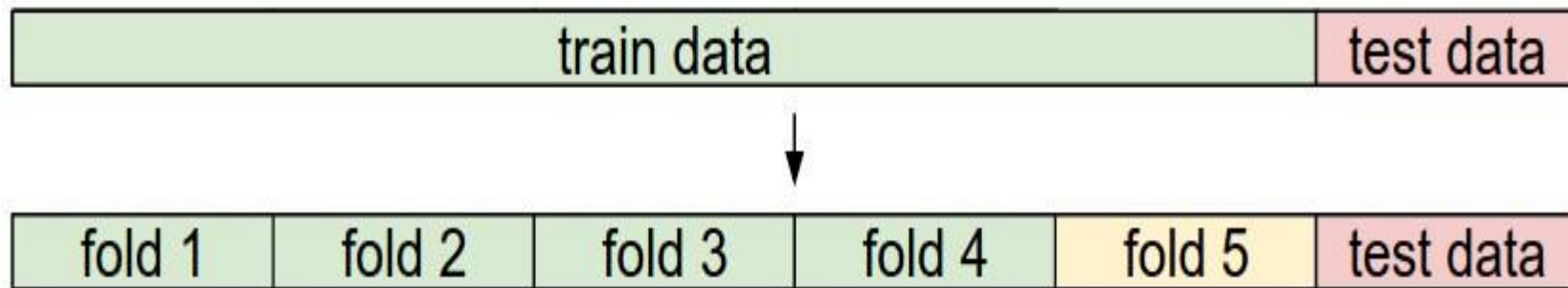
VALIDATION SET

- The idea is to split our training set in two: a slightly smaller training set, and what we call a **validation set**.



CROSS-VALIDATION

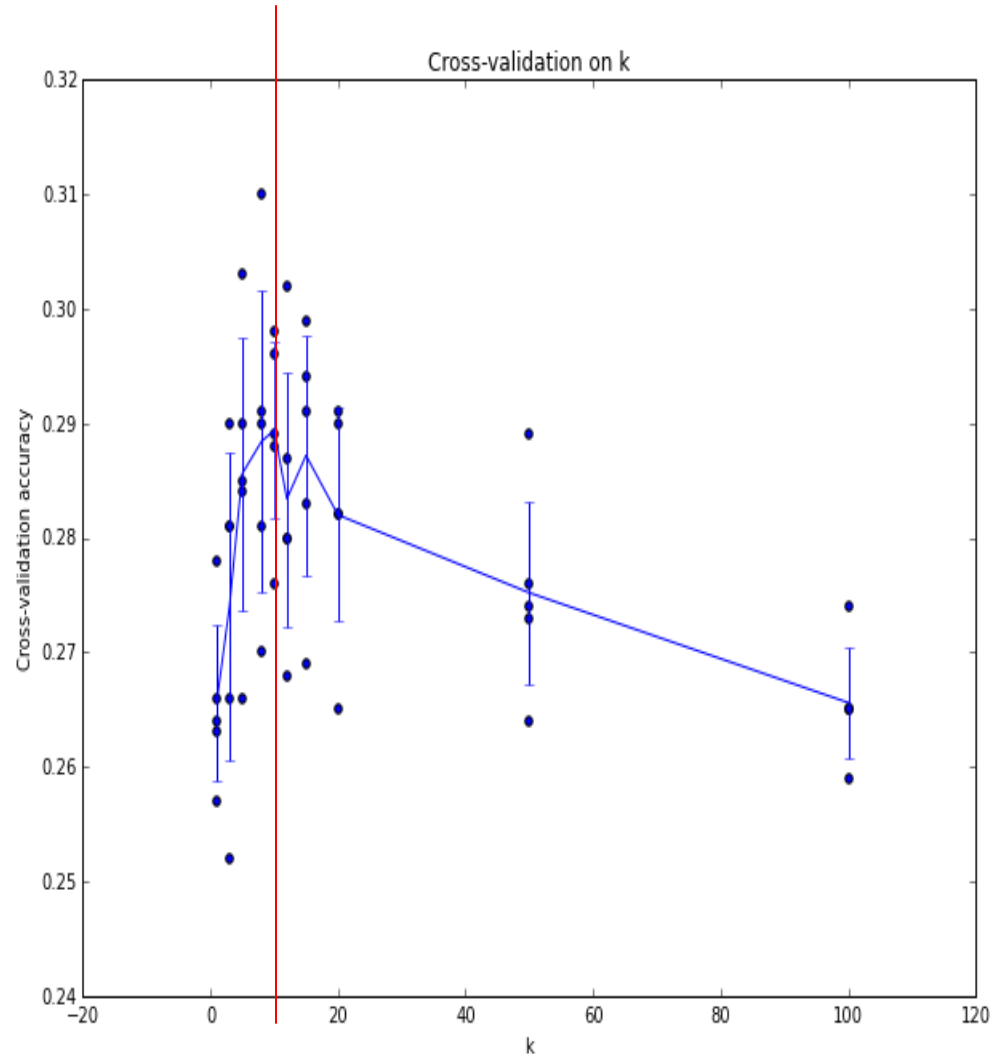
- In cases where the size of your training data might be small, we could use a technique for hyperparameter tuning called cross-validation.



Cross-validation

cycle through the choice of which fold is the validation fold, average results.

HYPERPARAMETER TUNING:



Example of 5-fold cross-validation for the value of k .

Each point: single outcome.

The line goes through the mean, bars indicated standard deviation

(Seems that $k \approx 7$ works best for this data)